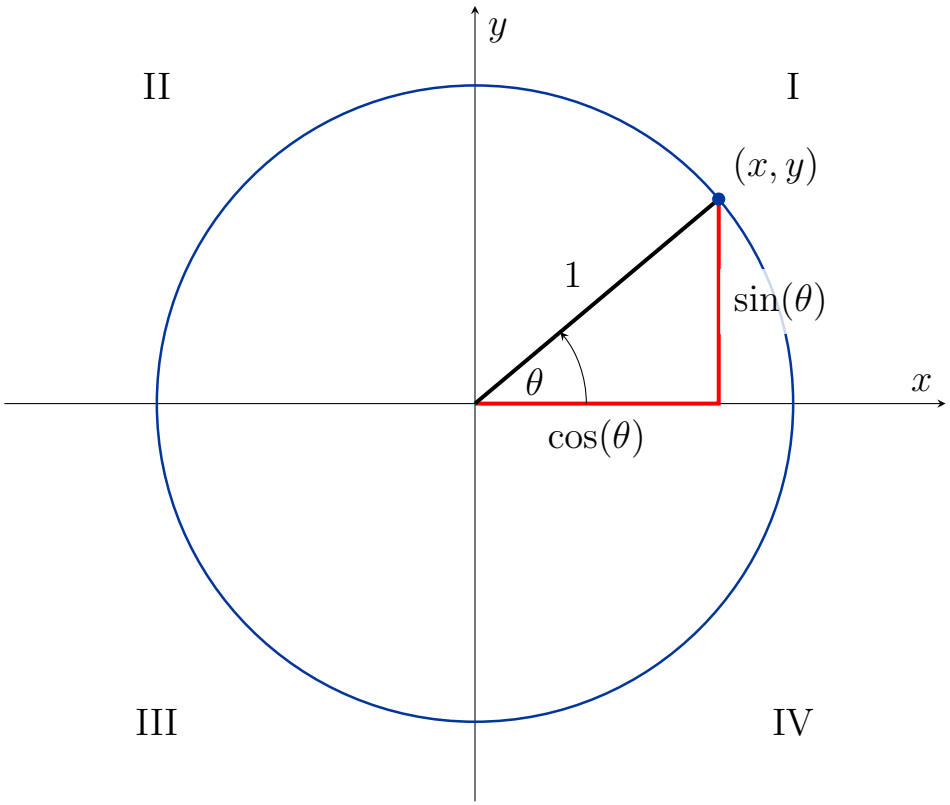
5.2: Unit Circle: Sine and Cosine Functions

Definition. (Unit Circle)
A **unit circle** has a center at $(0, 0)$ and radius 1. In a unit circle, the length of the intercepted arc is equal to the radian measure of the central angle θ.

Let (x, y) be the endpoint on the unit circle of an arc of arc length s.
The (x, y) coordinates of this point can be described as functions of the angle.



Definition. (Pythagorean Identity)
The **Pythagorean Identity** states that, for any real number θ,

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

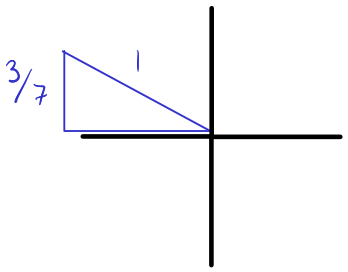
Example. If $\sin(\theta) = \frac{3}{7}$ and θ is in the second quadrant, find $\cos(\theta)$.

$\cos^2(\theta) + \sin^2(\theta) = 1$

$\cos^2(\theta) + \left(\frac{3}{7}\right)^2 = 1$

$\cos^2(\theta) = 1 - \frac{9}{49}$

$\cos(\theta) = \pm\sqrt{\frac{40}{49}} = \pm\frac{2\sqrt{10}}{7} \Rightarrow \boxed{\cos(\theta) = -\frac{2\sqrt{10}}{7}}$



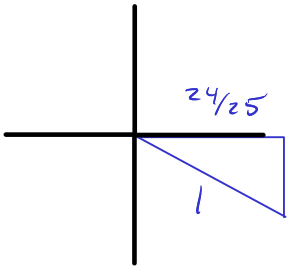
If $\cos(\theta) = \frac{24}{25}$ and θ is in the fourth quadrant, find $\sin(\theta)$.

$\cos^2(\theta) + \sin^2(\theta) = 1$

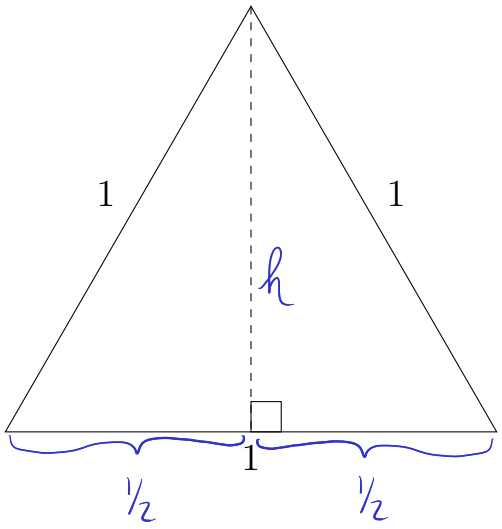
$\left(\frac{24}{25}\right)^2 + \sin^2(\theta) = 1$

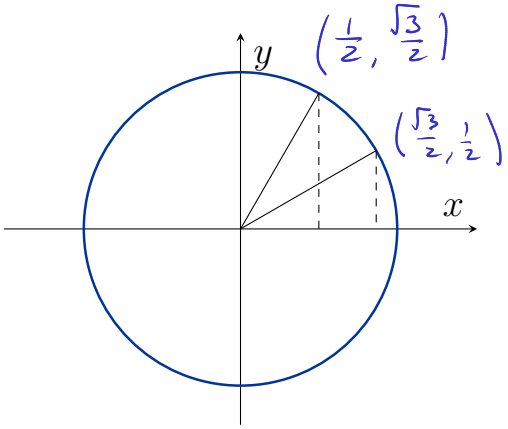
$\sin^2(\theta) = 1 - \frac{576}{625}$

$\sin(\theta) = \pm\sqrt{\frac{49}{625}} \rightarrow \boxed{\sin(\theta) = -\frac{7}{5}}$



We can find certain values of $\cos(\theta)$ and $\sin(\theta)$ using special triangles:



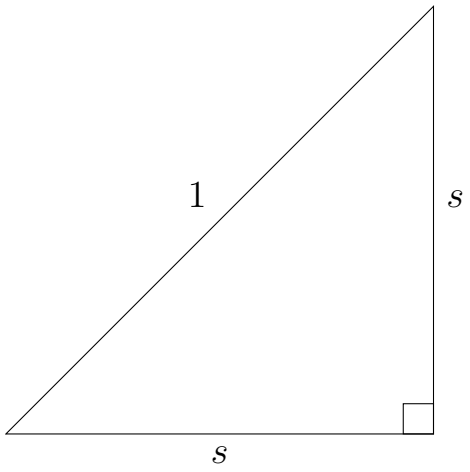


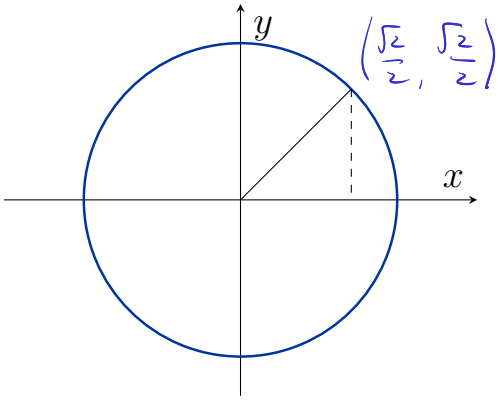
$$\left(\frac{1}{2}\right)^2 + h^2 = 1^2$$

$$h^2 = 1 - \frac{1}{4}$$

$$h = \sqrt{\frac{3}{4}} = \boxed{\frac{\sqrt{3}}{2}}$$

We can find certain values of $\cos(\theta)$ and $\sin(\theta)$ using special triangles:





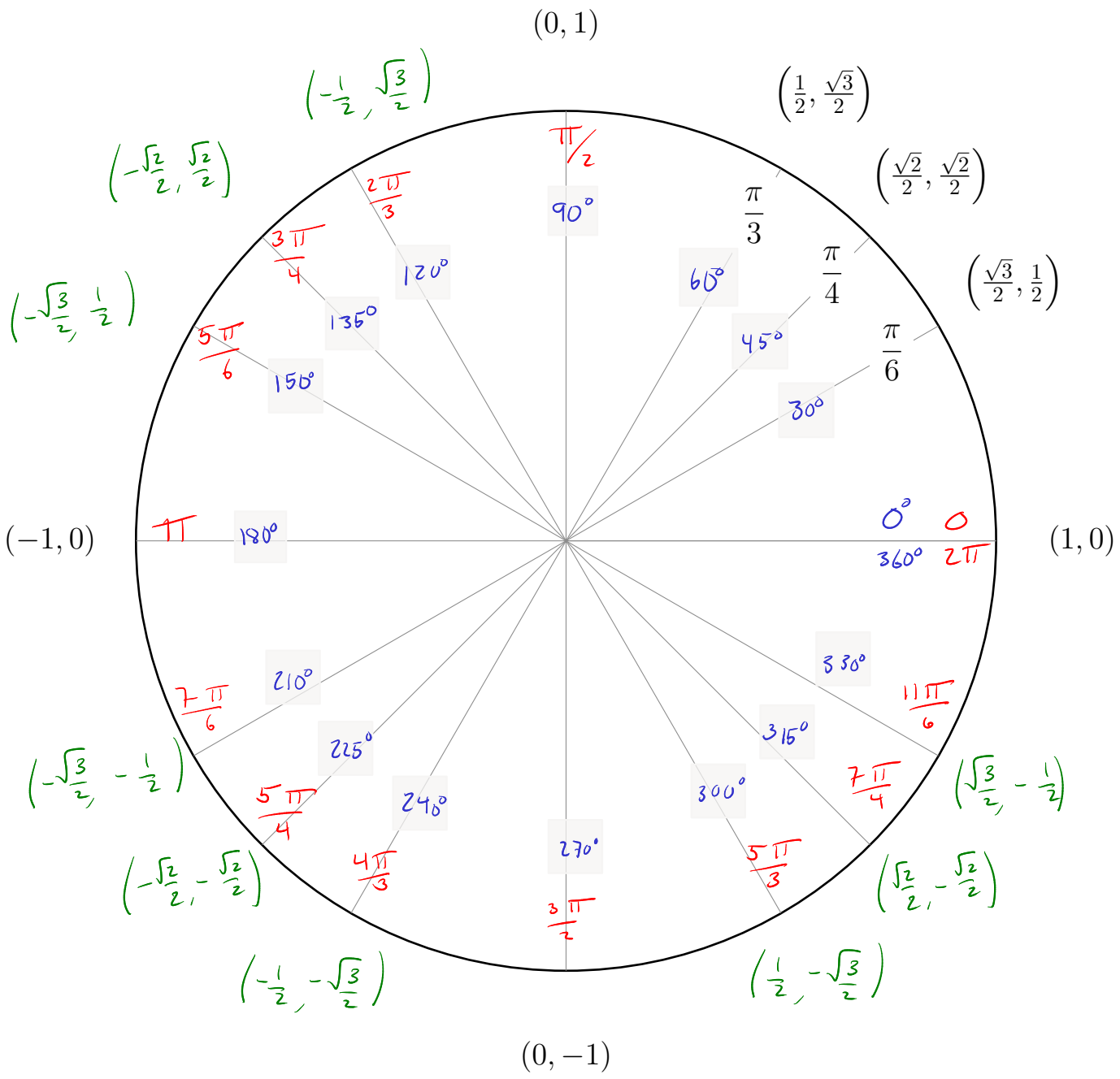
$s^2 + s^2 = 1$

$2s^2 = 1$

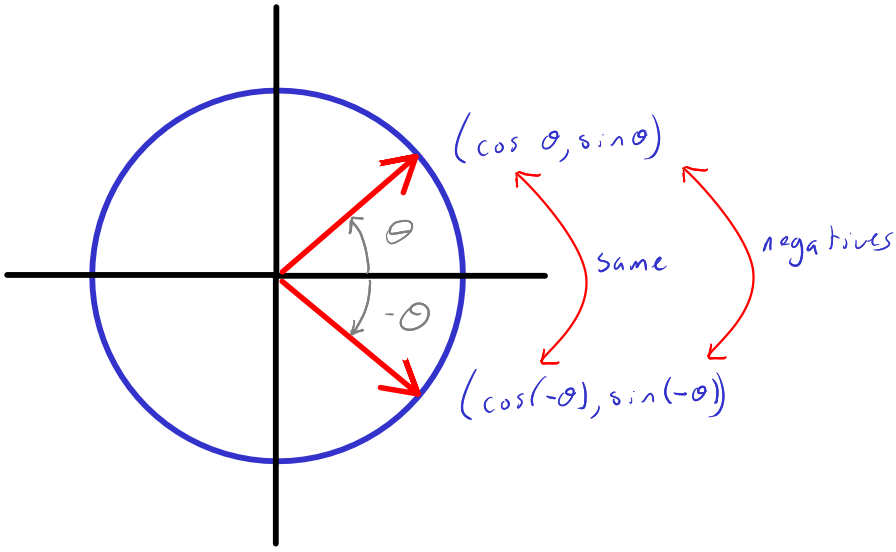
$s^2 = \frac{1}{2}$

$s = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \boxed{\frac{\sqrt{2}}{2}}$

Example. Fill out the remaining coordinates and radian measures below:



Example. Determine if $\cos(\theta)$ and $\sin(\theta)$ are even, odd, or neither.



$\cos(\theta)$ is even because

$$\cos(\theta) = \cos(-\theta)$$

$\sin(\theta)$ is odd because

$$-\sin(\theta) = \sin(-\theta)$$

Example. State the domain and range of $\cos(\theta)$ and $\sin(\theta)$

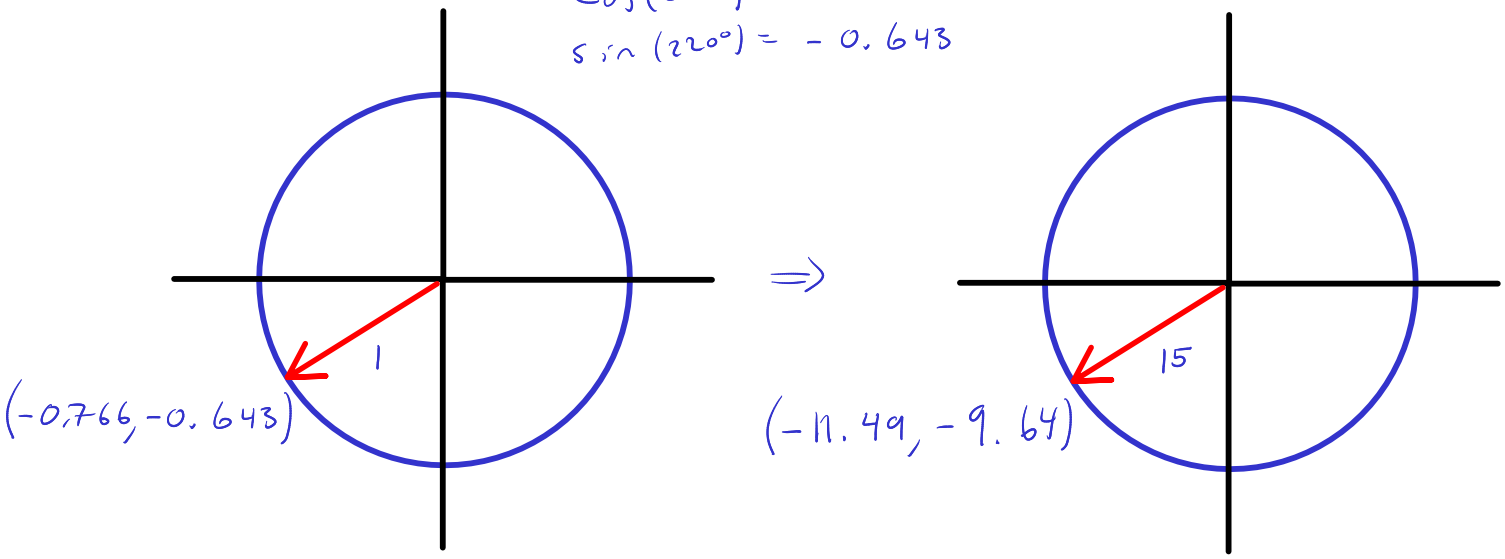
Domain: $(-\infty, \infty)$

Range: $[-1, 1]$

Example. Find the coordinate of the point on a circle with radius 15 corresponding to an angle of 220°.

$\cos(220°) = -0.766$

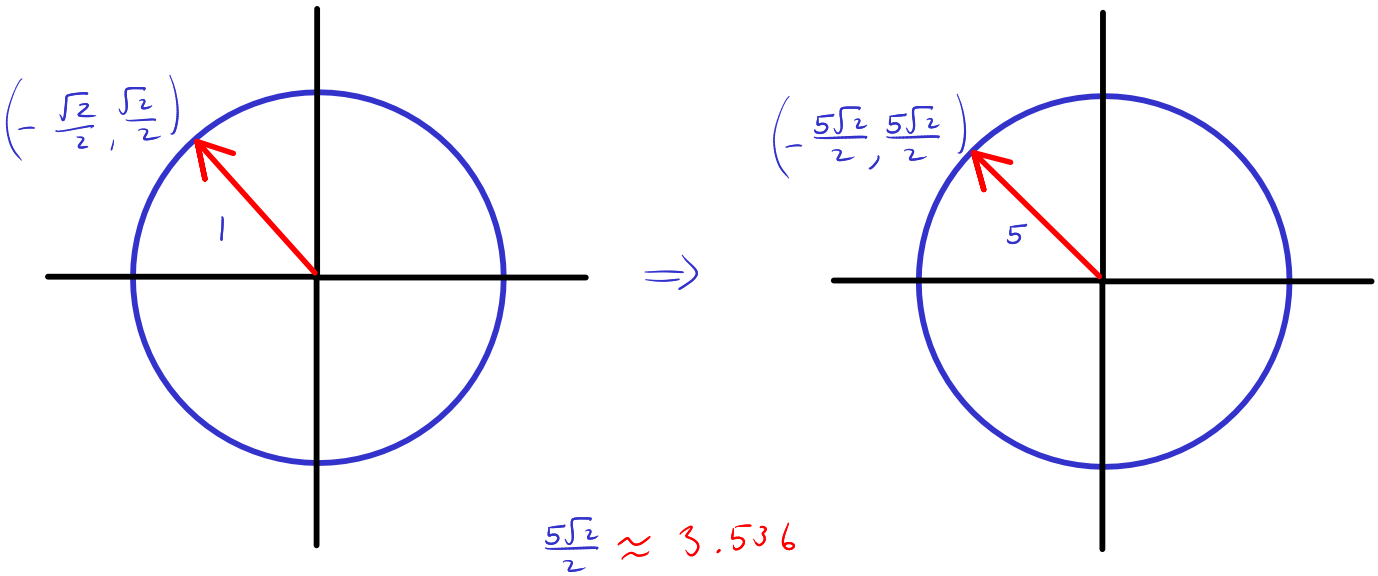
$\sin(220°) = -0.643$



Example. Find the coordinate of the point on a circle with radius 5 corresponding to an angle of $\frac{3\pi}{4}$.

$\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$\sin\left(\frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2}$



$\frac{5\sqrt{2}}{2} \approx 3.536$